

Geometry
Unit 9
Lesson 9 Practice

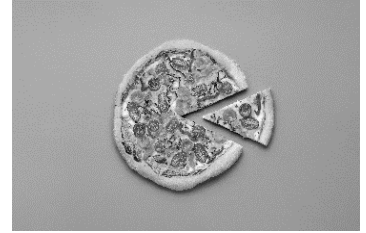
Name_____ **Answer Key** _____
Date_____
Period_____

1. Rylan enjoys adding her own toppings to a frozen pizza. She bakes her frozen pizza on a pizza stone that is 12 inches in diameter and $\frac{1}{4}$ inch thick. What is the surface area of the pizza stone? Round to the nearest hundredth.

$$r = 6$$

$$h = \frac{1}{4}$$

$$SA = 2\pi r^2 + 2\pi rh = 2\pi(6)^2 + 2\pi(6)(0.25) \approx 235.62 \text{ in}^2 \text{ (when using the pi button)}$$



2. Rylan covers every possible space on top of the pizza with her favorite toppings. She only adds toppings to the top of a 12 inch diameter pizza. What is the area she has for her various toppings? Round to the nearest hundredth.

Area of circle

$$A = \pi r^2 = \pi(6)^2 \approx 113.1 \text{ in}^2 \text{ (when using the pi button)}$$

3. A ping pong ball has a diameter of 40 millimeters. What is the surface area of a ping pong ball? Round to the nearest hundredth.

Sphere

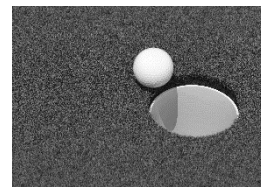
$$r = 20$$

$$SA = 4\pi r^2 = 4\pi(20)^2 \approx 5026.55 \text{ mm}^2 \text{ (when using the pi button)}$$

4. The minimum diameter of a golf ball is 42.67 millimeters. What is the minimum surface area of a golf ball? Round to the nearest hundredth.

$$r = 21.335$$

$$SA = 4\pi r^2 = 4\pi(21.335)^2 \approx 5719.99 \text{ mm}^2 \text{ (when using the pi button)}$$



5. What is the difference in surface area between a golf ball and a ping pong ball?
 $5719.99 - 5026.55 = 693.44 \text{ mm}^2$

6. Harry's barn needs a new coat of paint. The stable area of the barn is a rectangular prism with a base of 40 feet by 60 feet and a height of 12 feet. The loft area of the barn is a triangular prism laying on one side above the stable area. The other two sides of the triangular prism are comprised of roof shingles, so the only portion that needs paint are the two base triangles. The height of the loft is 10 feet, with a base width of 40 feet. Find the area of the barn that needs to be painted.



Lateral area of rectangular prism plus two triangles

$$(200)(12) + \frac{1}{2}(10)(40)(2) = 2400 + 400 = 2800 \text{ ft}^2$$

7. If 1 gallon of paint covers 400 square feet, how much paint does Harry need to buy?

$$\frac{2800}{400} = 7 \text{ gallons of paint}$$

8. If 1 gallon of paint costs \$20.90, how much will the paint cost?

$$7(20.90) = \$146.30$$

9. Bert wants to build a new pen for his chickens. He has one area of his property available for a rectangular pen that would measure 20 feet by 35 feet. He has a second area of his property available for a square pen that would measure 27 feet wide. Which pen area would require the least amount of fence?

$$\text{Rectangular fence: } 2(20 + 35) = 110 \text{ ft}$$

$$\text{Square fence: } (27)(4) = 108 \text{ ft}$$

The square fence would require the least amount of fence.



10. Bert decides to build the pen area with the least amount of fence. If his fence is 4.5 feet tall, what is the surface area of his fence?

$$Ph = (108)(4.5) = 486 \text{ ft}^2$$